NAME:VARAD PATIL **C1 CAT 2 PAPER**
REGISTER NUMBER: 24BIT0175

b) An 8051-based system uses an 11.0592 MHz crystal. An LED is connected to P1.0. Write a program using Timer 0 in Mode 1 and a timer interrupt to toggle the LED every 10 ms. Your program should:

- Configure Timer 0.
- Calculate the initial values of TH0 and TL0.
- Enable Timer 0 interrupt.
- Enable global interrupts.
- Reload the timer inside the ISR.
- Toggle P1.0 after the required number of interrupts.

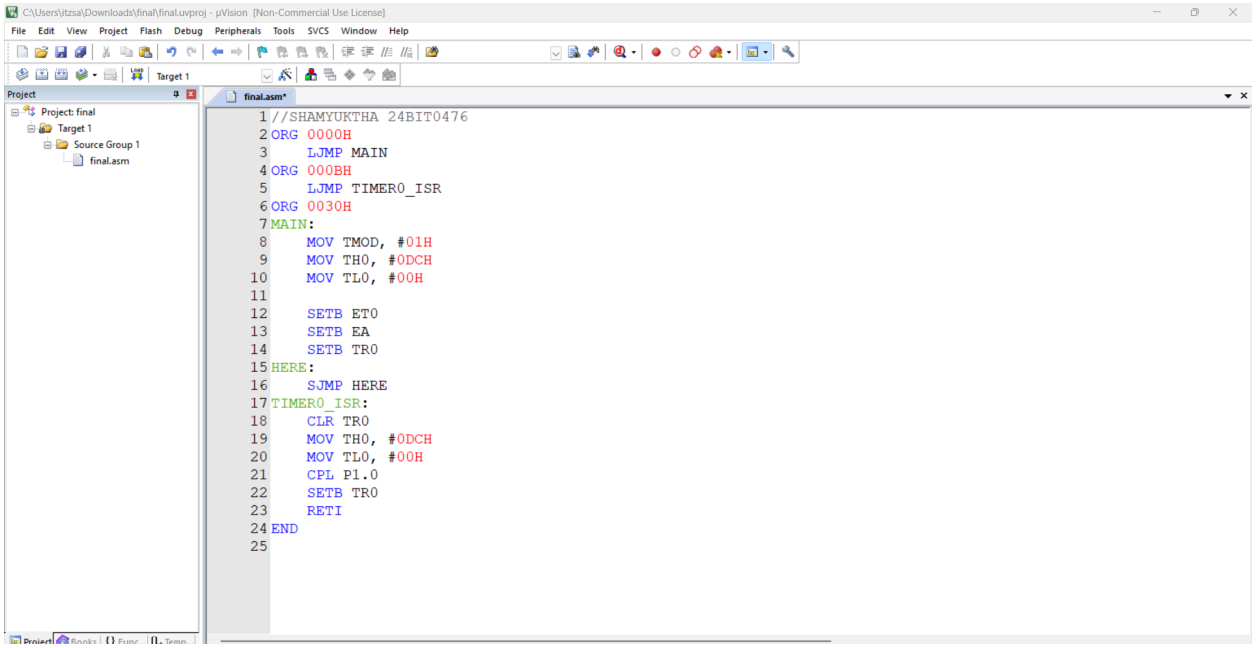
Show the calculations clearly

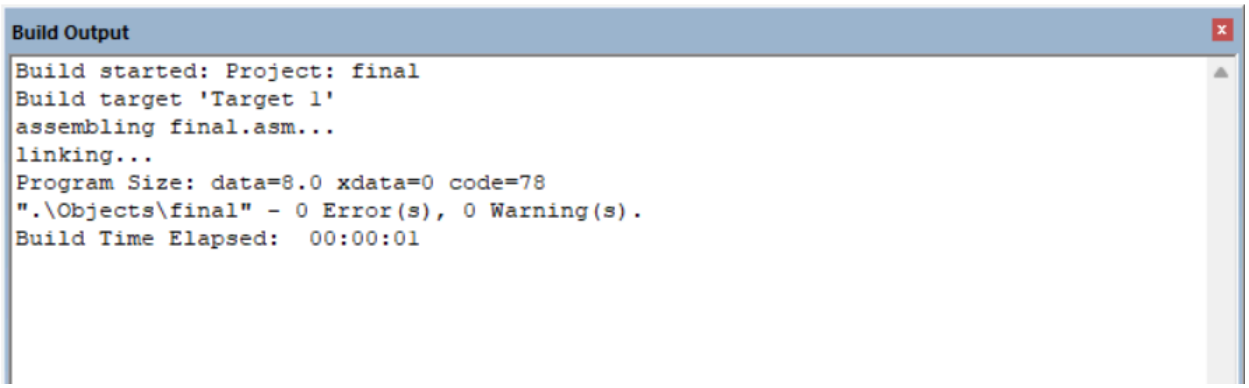
1. Calculations

Step	Working
Machine cycle time	12 / 11.0592 MHz = 1.0851 µs (timer clock = 11.0592 MHz / 12 = 921.6 kHz)
Counts needed for 10 ms	10 ms / 1.0851 µs = $10 \times 10^{-3} \times 921\,600 = 9216$ counts
Initial count (Mode 1, 16-bit)	65536 – 9216 = 56320 (decimal)
Convert to hex	56320 = 0xDC00
Initial values	TH0 = DCH , TL0 = 00H
TMOD value	Timer 0, Mode 1, timer (C/T = 0), GATE = 0 → TMOD = 0000 0001b = 01H
Interrupts per toggle	Timer overflow already occurs every 10 ms, so 1 interrupt = 1 toggle. LED period = 20 ms (50 Hz).

OUTPUT:

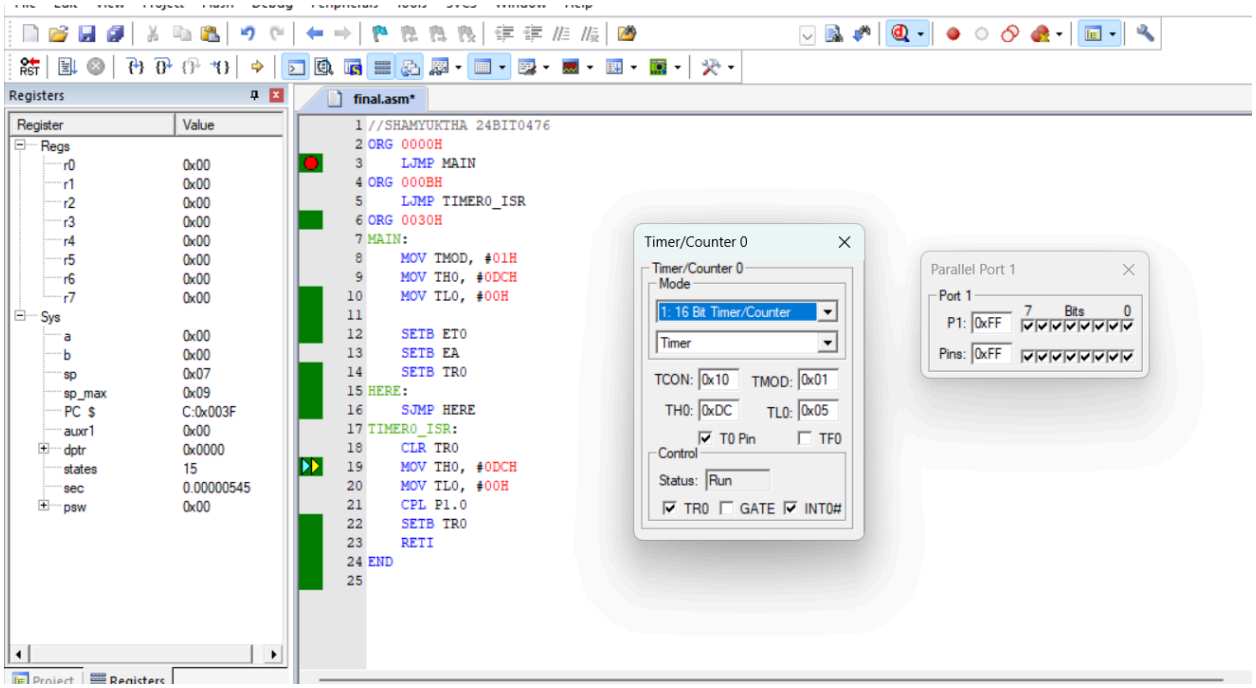
Source code and successful build:



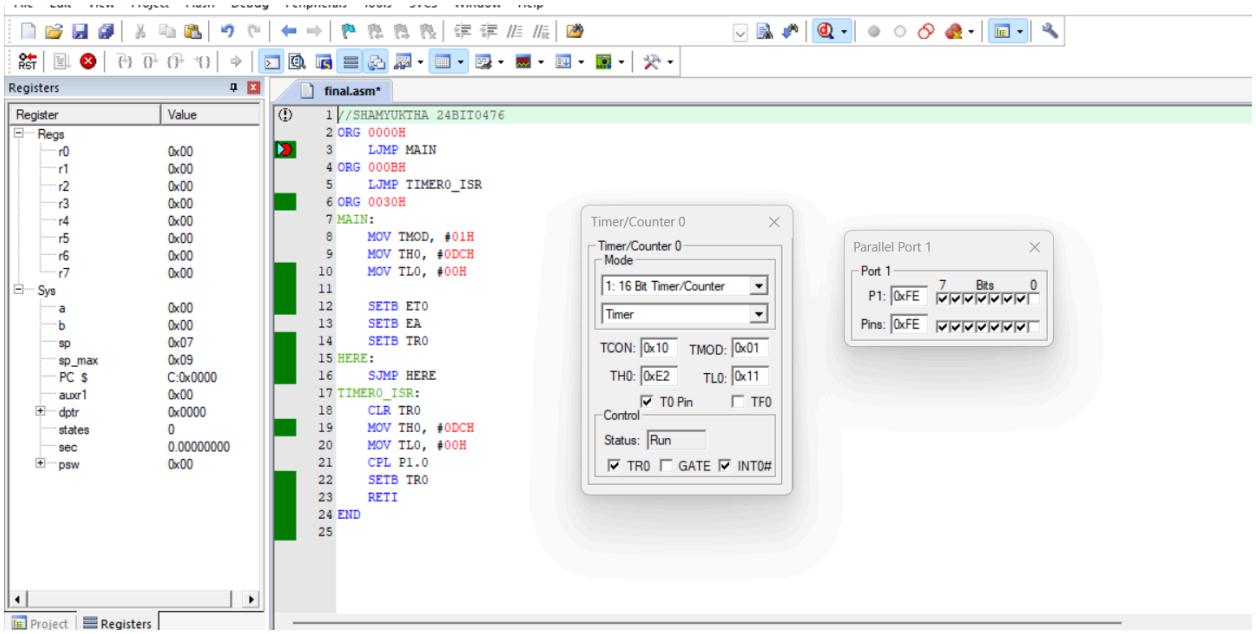


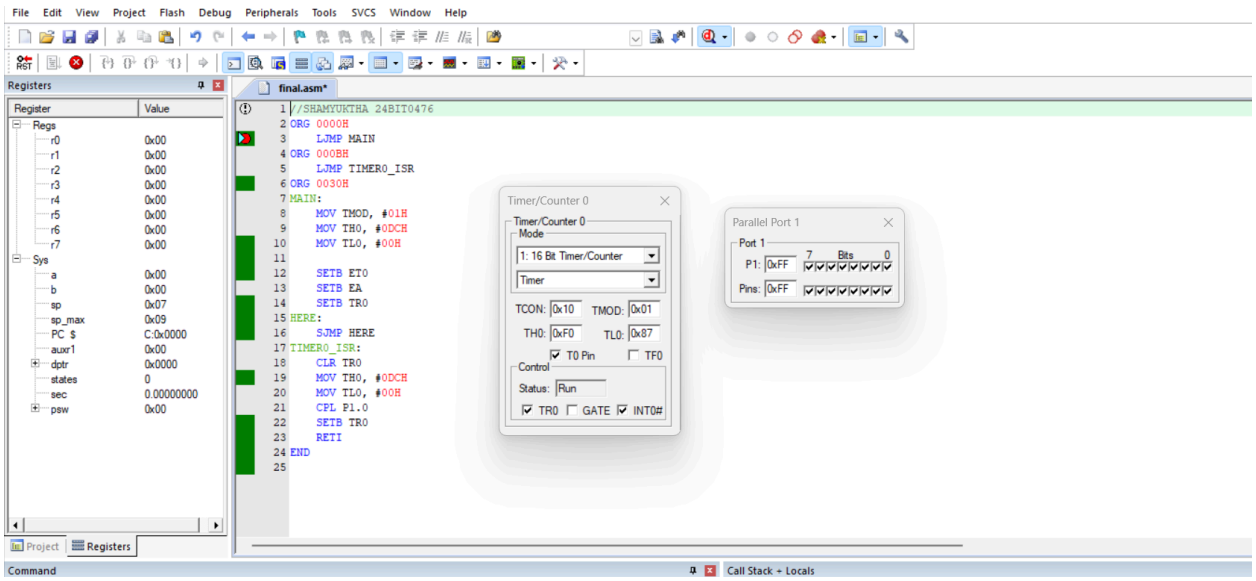
DEBUG:

Timer 0 loaded with initial values (before first overflow)



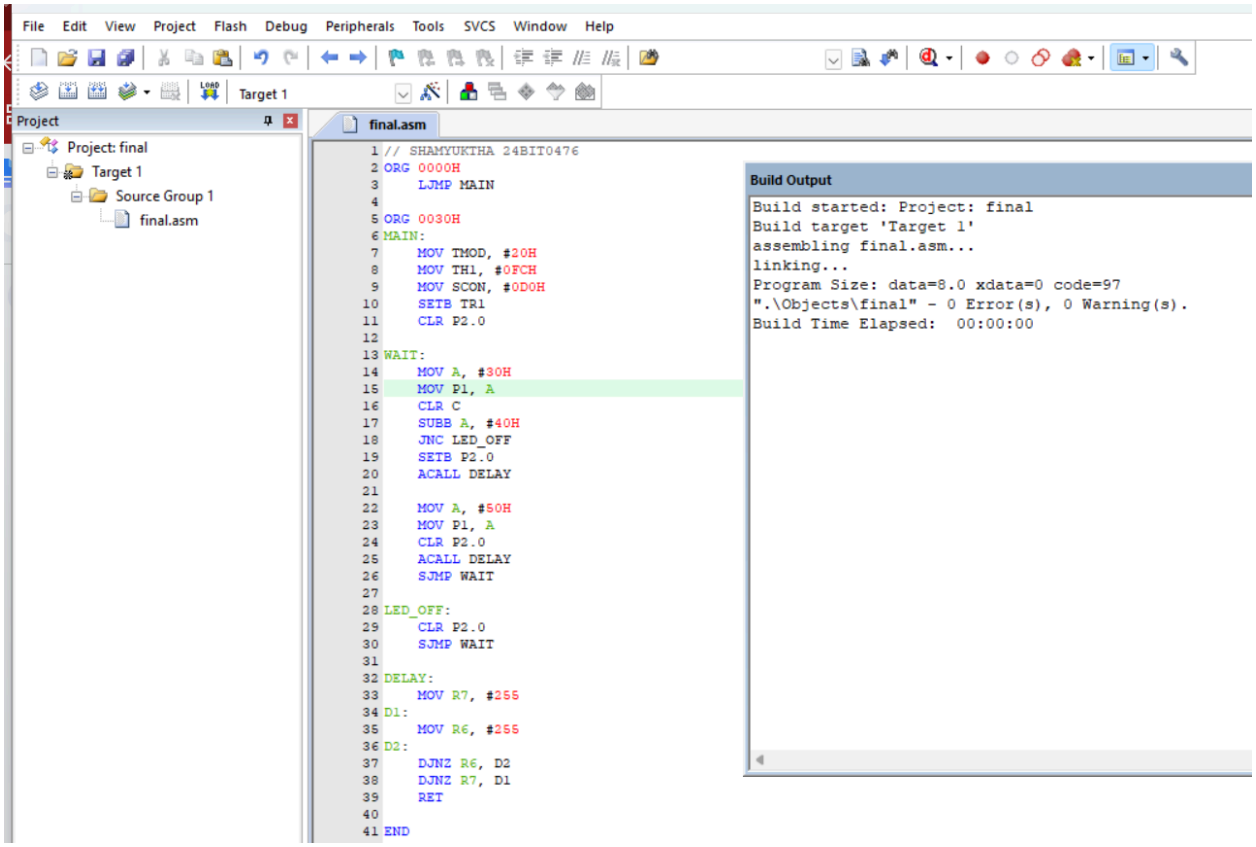
: After the 1st interrupt – P1.0 toggled



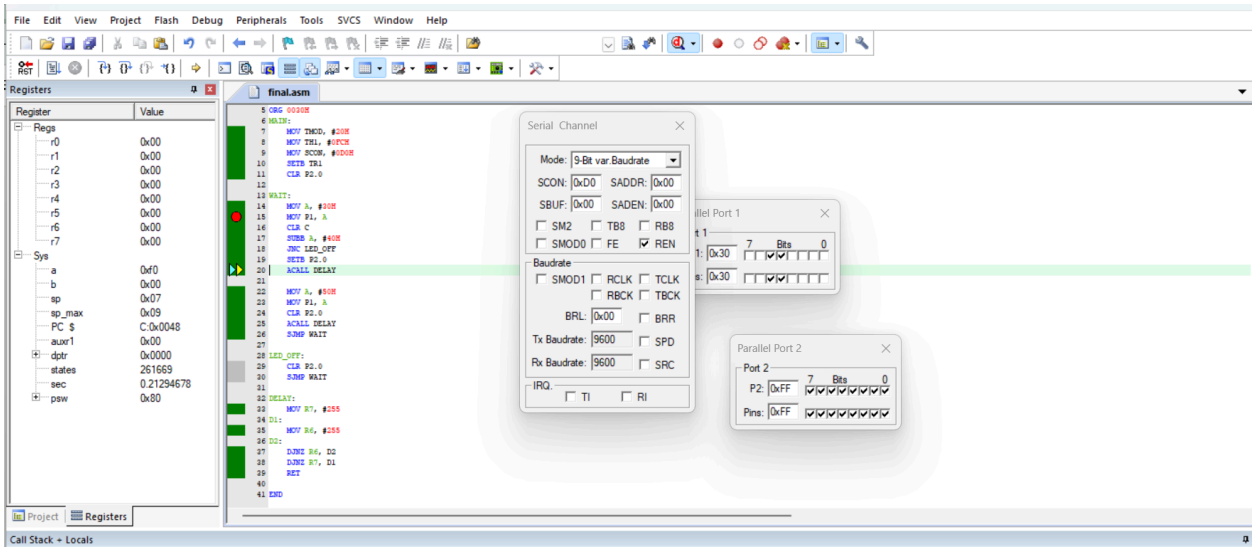


5.	A smart irrigation controller communicates with four field monitoring units over a common serial communication bus. Each field unit transmits an 8-bit soil-moisture reading along with a 9th bit used to distinguish between address/control information and ordinary sensor data. The central controller is based on an 8051 microcontroller with a 14.7456 MHz crystal. The communication system must operate at a baud rate of 9600 bps. Develop an 8051 assembly language program for the central controller with the following requirements: • Configure the UART for 9-bit communication and enable multiprocessor communication. • Continuously monitor the common serial bus for incoming frames. • Transfer the received 9th bit (RB8) to the Carry flag. • If the 9th bit is 0, treat the received byte as a soil-moisture data frame and display the 8-bit value on Port P1. • If the 9th bit is 1, treat the received byte as an address/control frame and do not display it on P1. • If a valid soil-moisture value is below 40H, turn ON an irrigation indicator LED connected to P2.0; otherwise, turn it OFF. Continue receiving and processing subsequent frames indefinitely.	10	2

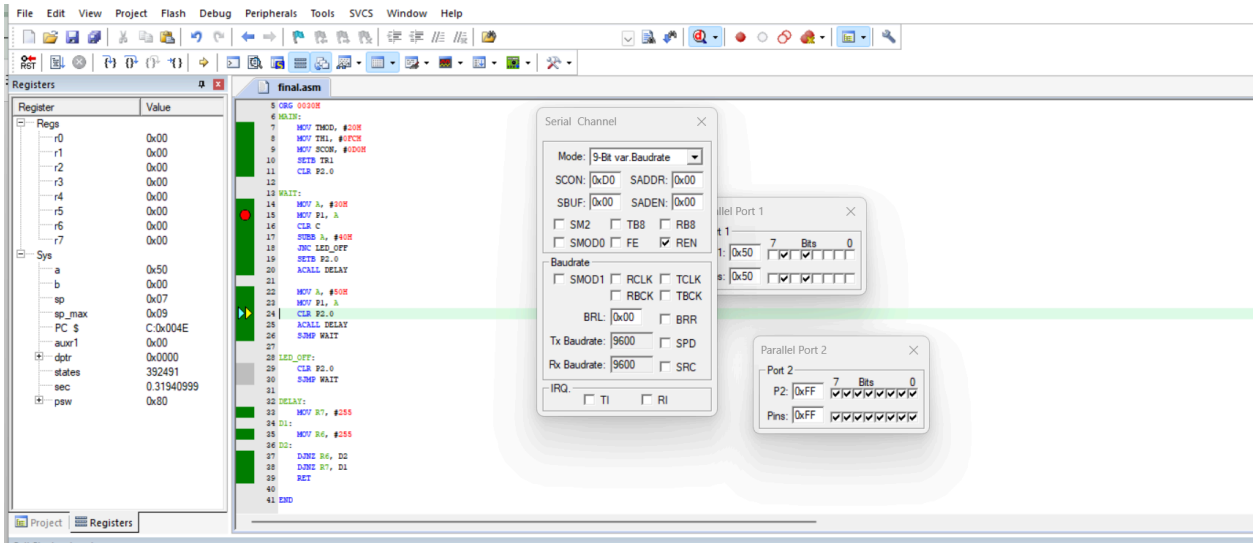
Source code and successful build:



INPUT IS 30 SO LED IS ON



LED OFF WHEN INPUT GREATER THAAN 40



When input greater than 40

